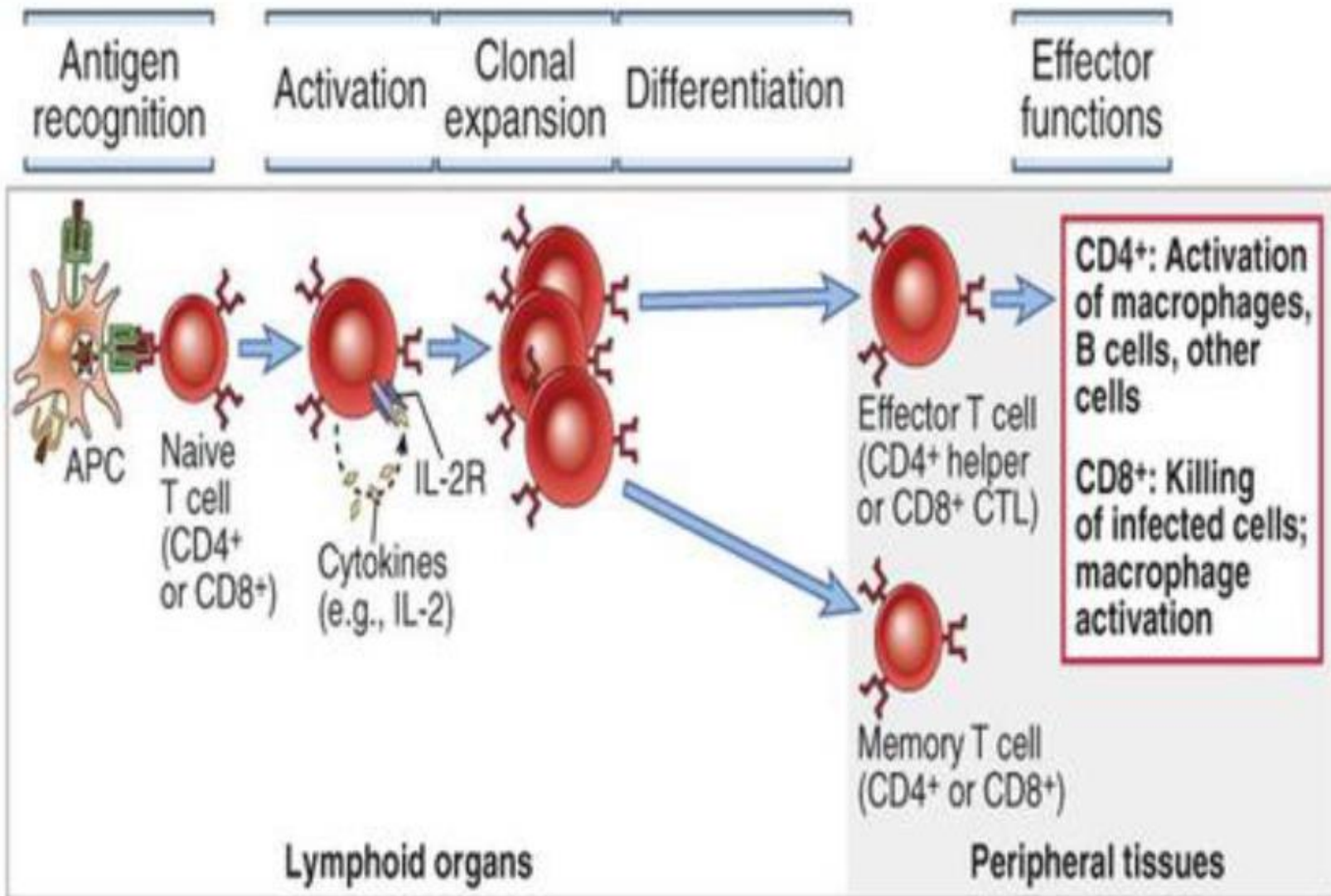


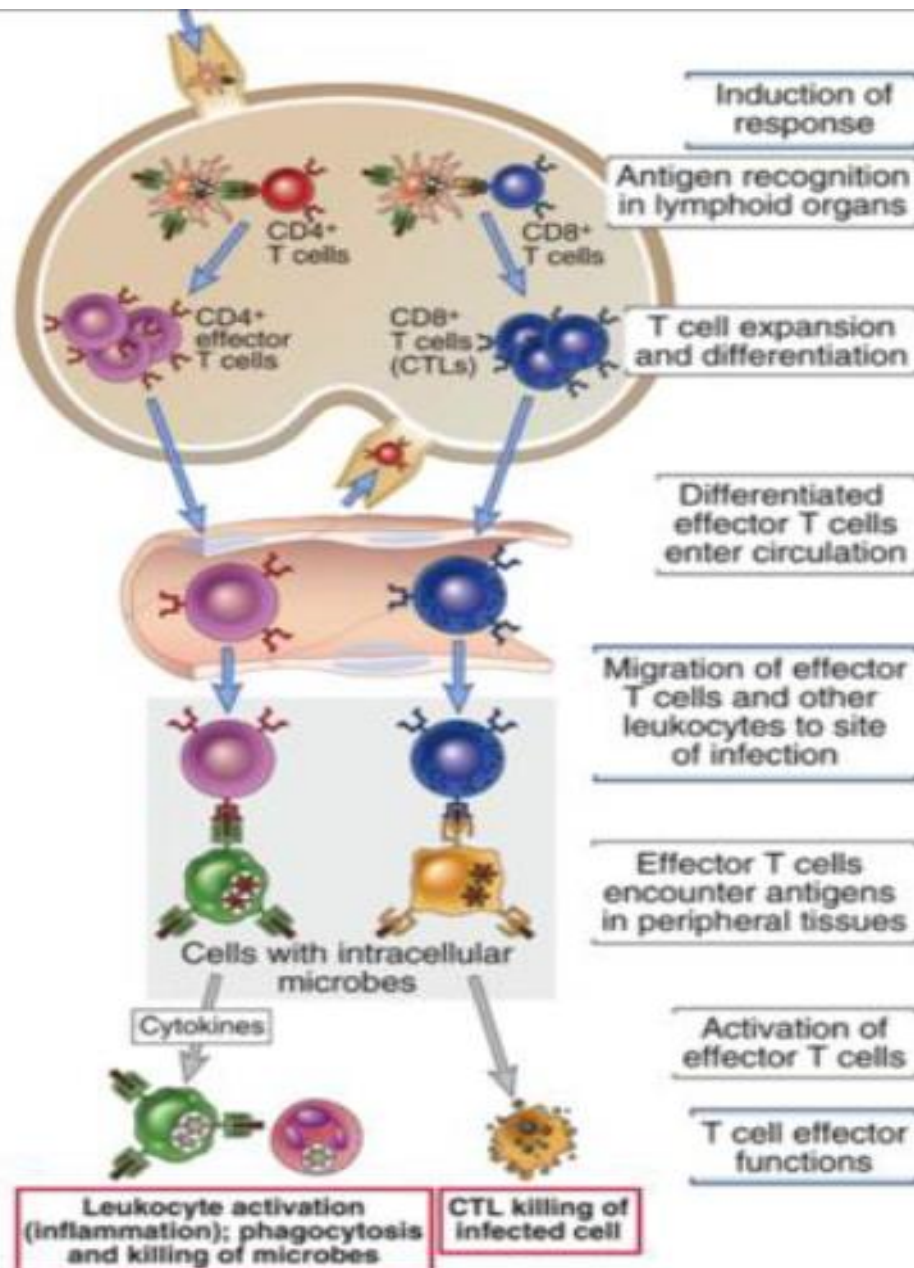
Cell Mediated Immune Response

Cell Mediated Immunity

- Provided by T lymphocytes
- Provides immunity to (i) intracellular bacteria (ii) viruses, (iii) fungi, (iv) protozoa and (v) tumours
- T cells can recognize antigen only when it is presented on the surface of Antigen presenting cells (APCs) by self MHC molecules
- This self MHC restriction results from positive selection during maturation of T cells in the thymus

Phases of T cell responses



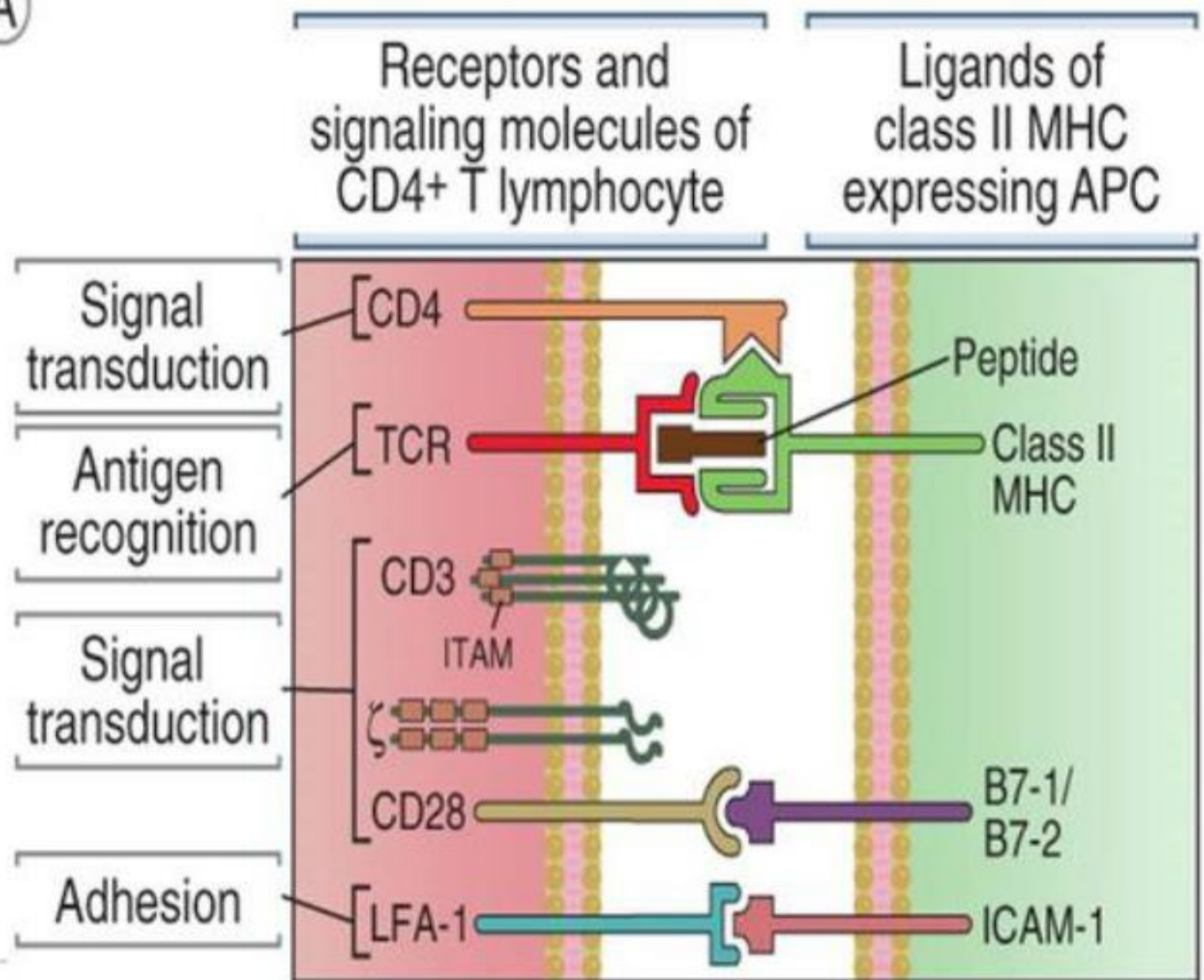


Mechanism of CMI

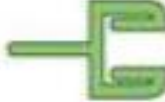
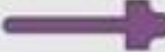
Antigen recognition by T cells

- Initiation of T cell responses requires multiple receptors on T cells
- recognizing ligands on APCs,
 - » TCR=MHC associated peptides
 - » CD4 or CD8 co-receptors = MHC molecules
 - » Adhesion molecules = strengthen the binding of T cells to APCs
 - » Receptors for co-stimulators recognize = second signals provided by the APCs

A

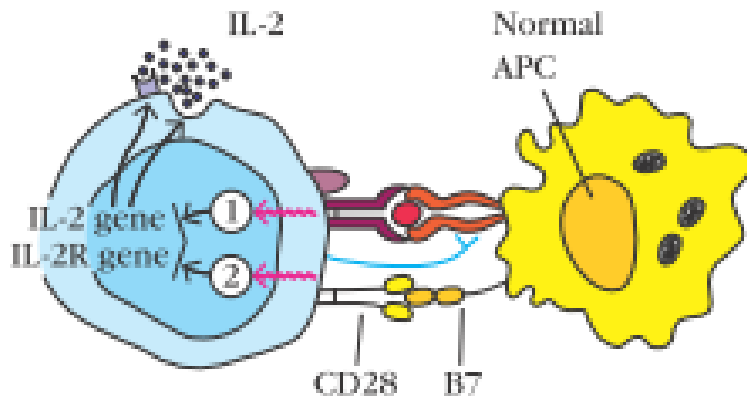


B

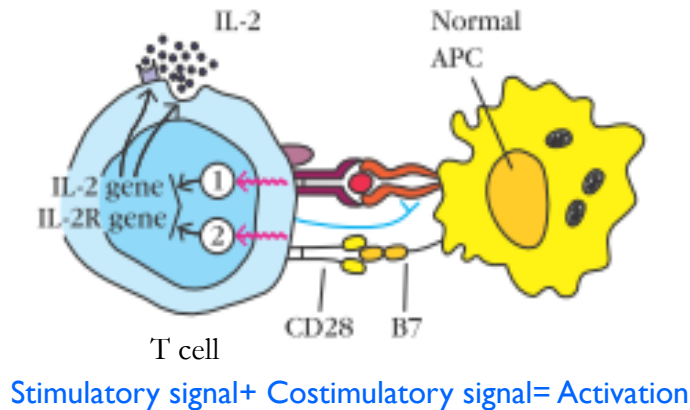
Surface molecules of T lymphocytes	Function	Ligand	
		Name	Expressed on
CD3 	Signal transduction by TCR complex	None	
	Signal transduction by TCR complex	None	
CD4 	Signal transduction	Class II MHC 	Antigen-presenting cells
CD8 	Signal transduction	Class I MHC 	All nucleated cells
CD28 	Signal transduction (costimulation)	B7-1/B7-2 	Antigen-presenting cells
CTLA-4 	Negative regulation	B7-1/B7-2 	Antigen-presenting cells
LFA-1 	Adhesion, signal transduction	ICAM-1 	Antigen-presenting cells, endothelium
VLA-4 	Adhesion, signal transduction	VCAM-1 	Endothelium

Mechanism of CMI

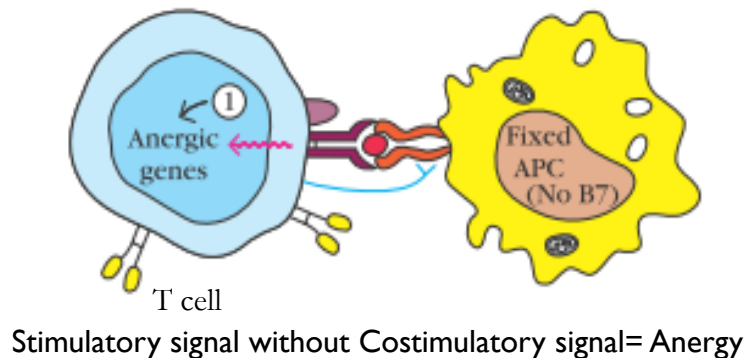
- Antigen recognition by TCR provides Stimulatory signal (Signal 1) to the T cell
- Binding of B7 molecule on APC with CD28 molecule on T cell provides Co-stimulatory signal (Signal 2) to the T cell



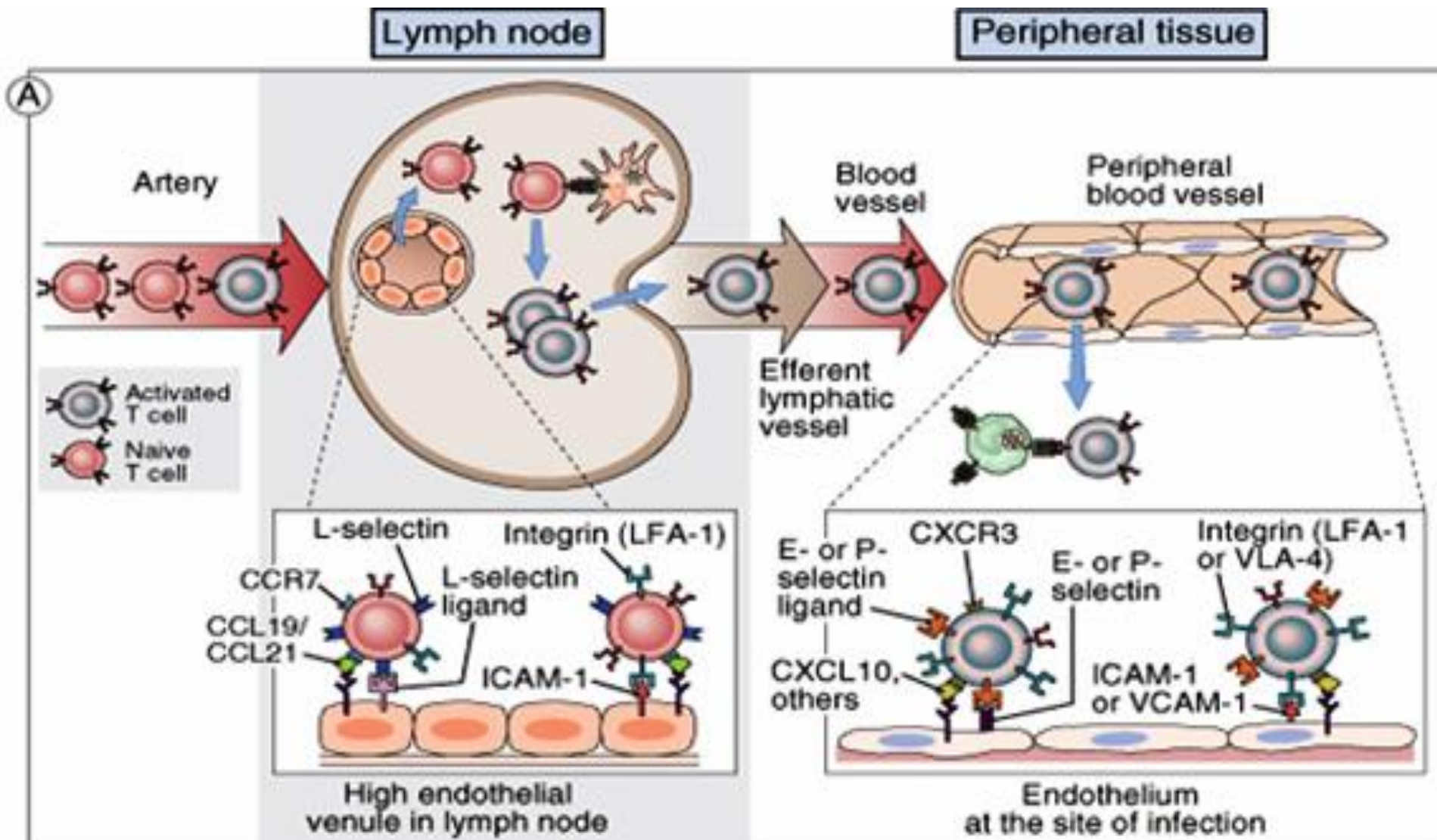
T cell activation



- When T cells receive both Stimulatory (Signal 1) and Co-stimulatory (Signal 2) signals they are activated
- If the T cells receive only the Stimulatory signal without Co-stimulatory signal, they are permanently inactivated (clonal anergy)



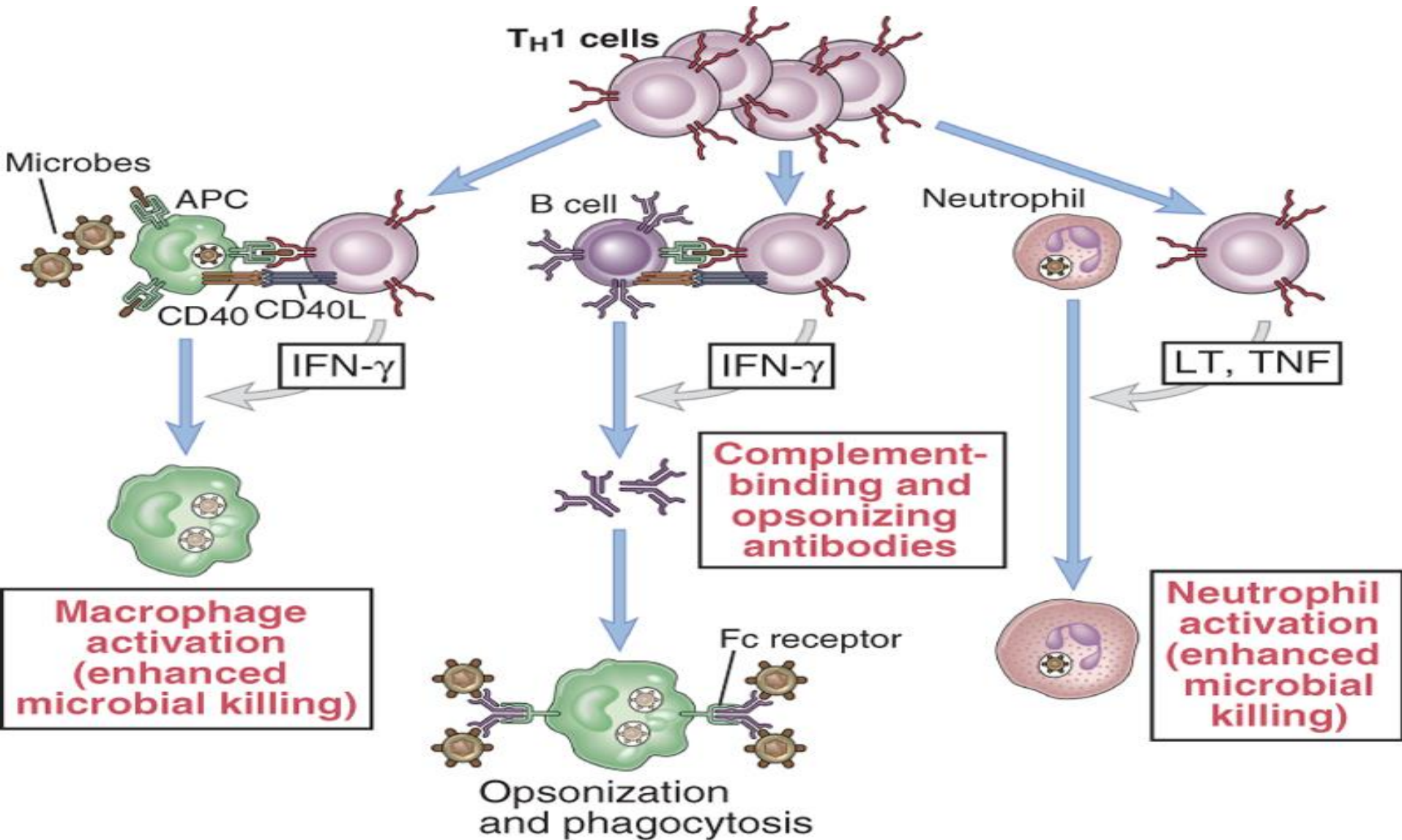
Migration of naïve and effector T lymphocytes



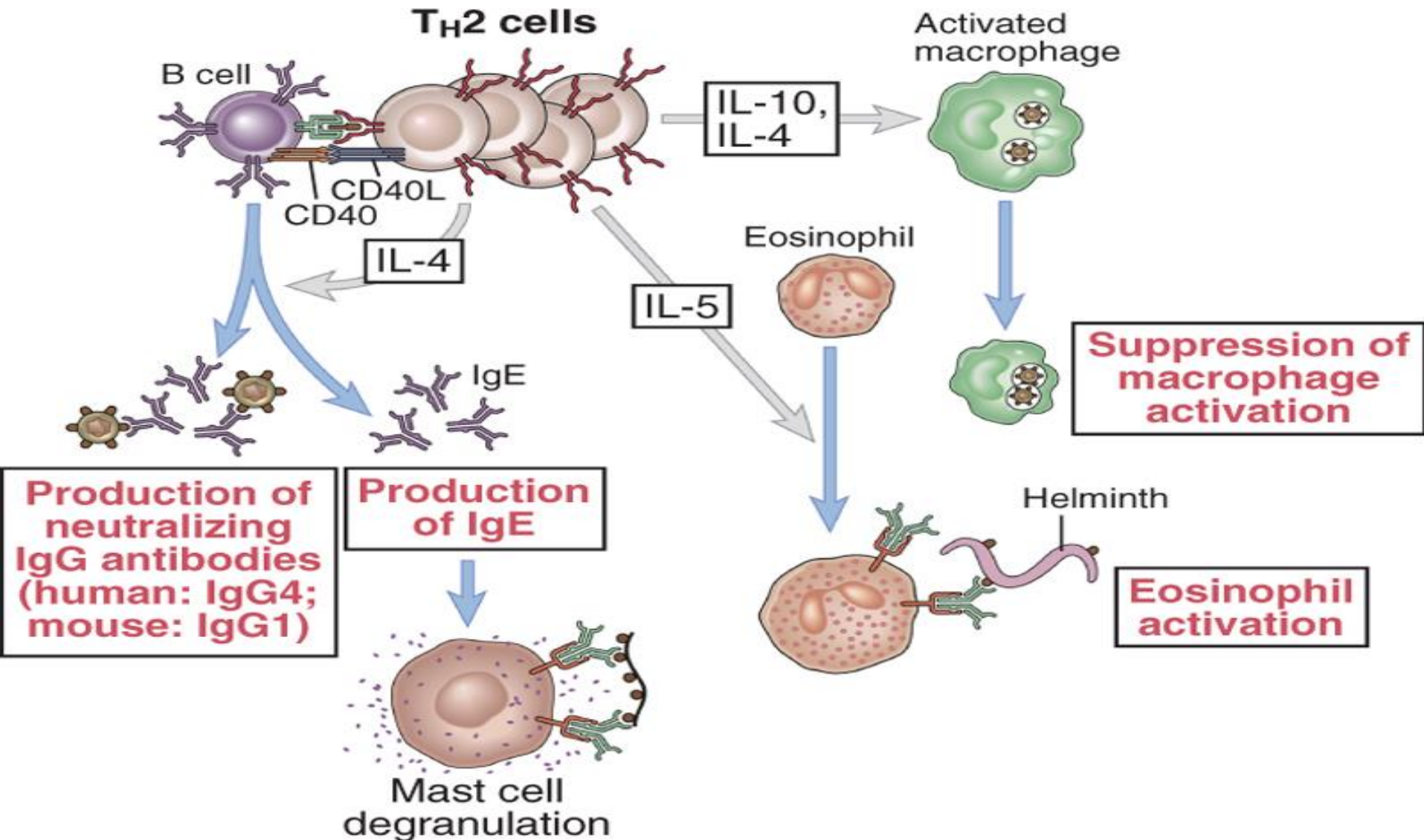
T cell homing receptors and their ligands

B T cell molecules involved in homing	Endothelial cell molecules	Function of receptor: ligand pair
Naive T cells  L-selectin  LFA-1 ($\beta 2$-integrin)  CCR7	 L-selectin ligand  ICAM-1  CCL19 or CCL21	Adhesion of naive T cells to high endothelial venule (HEV) in lymph node Stable arrest on HEV Activation of integrins and chemotaxis
Activated (effector and memory) T cells  E- and P-selectin ligand  LFA-1 ($\beta 2$-integrin) or VLA-4 ($\beta 1$ integrin)  CXCR3, others	 E- or P-selectin  ICAM-1 or VCAM-1  CXCL10, others	Initial weak adhesion of effector and memory T cells to cytokine activated endothelium at peripheral site of infection Stable arrest on cytokine activated endothelium at peripheral site of infection Activation of integrins and chemotaxis

Effector function of Th1 cells



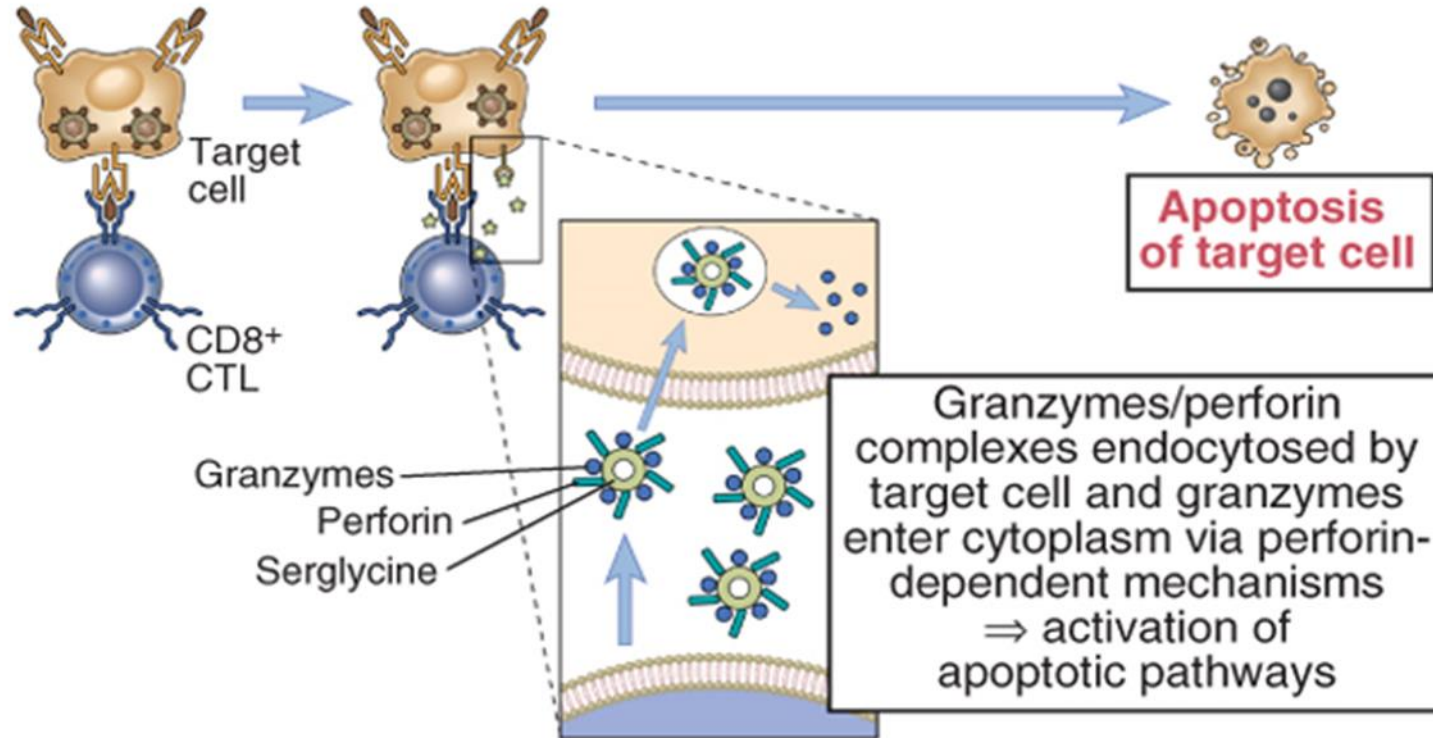
Effector function of Th2 cells



Mechanisms of CTL-mediated lysis of target cells

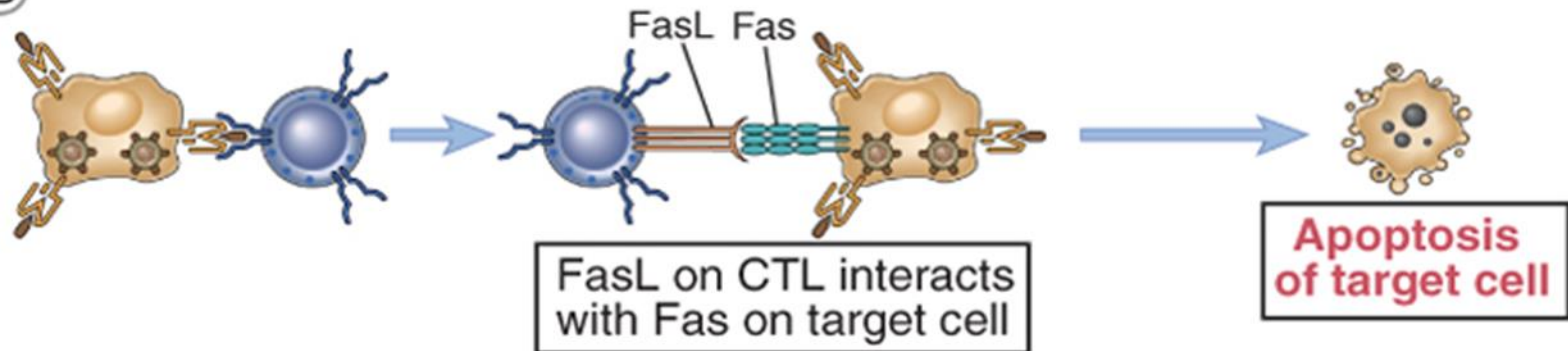
A

Granule exocytosis $\Rightarrow$
entry of
granzymes $\Rightarrow$
activation of
caspases $\Rightarrow$
apoptosis of
target cell

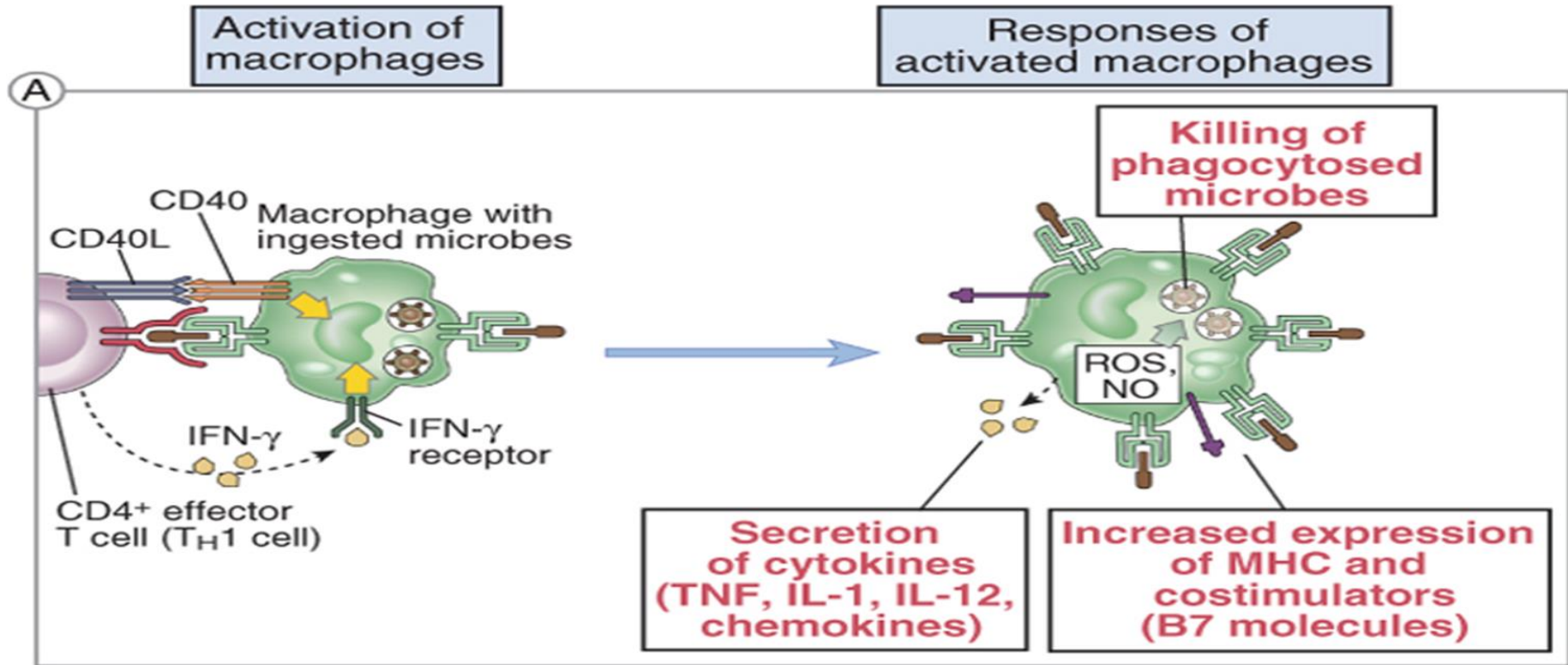


B

FasL-Fas-mediated
target cell
apoptosis



Action and functions of macrophages in CMI



B

Macrophage response	Role in cell-mediated immunity
Production of reactive oxygen species, nitric oxide, increased lysosomal enzymes	Killing of microbes in phagolysosomes (effector function of macrophages)
Secretion of cytokines (TNF, IL-1, IL-12)	TNF, IL-1: leukocyte recruitment (inflammation) IL-12: T _H 1 differentiation, IFN- γ production
Increased expression of B7 costimulators, MHC molecules	Increased T cell activation (amplification)

Functions of activated macrophages in CMI

Macrophage response

Role in CMI

Production of reactive oxygen species, nitric oxide, lysosomal enzymes

Killing of microbes in phagolysosomes (main effector function)

Secretion of cytokines (TNF, IL-1, chemokines, others)

Leukocyte recruitment (inflammation)

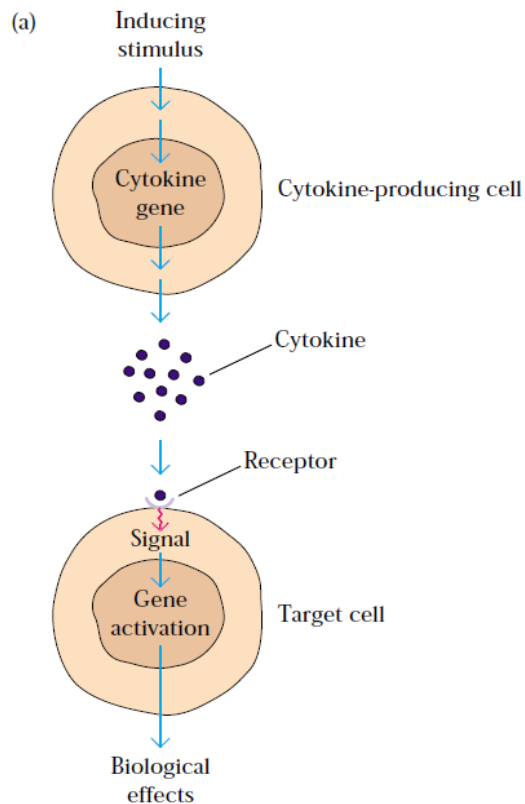
Increased expression of B7 costimulators, MHC molecules

Increased T cell activation (adaptive immunity)

Macrophages are activated by TLR recognition of microbes as part of innate immunity; T cells activate them more

Cytokines

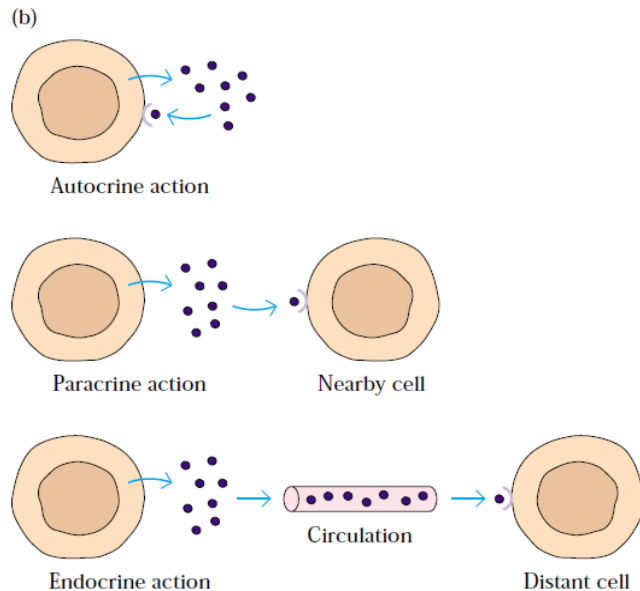
- ◆ Protein molecules secreted by cells that regulate function of that cell or other cells
- ▶ The name denotes their role in cell to cell communication
- ▶ Development of an effective immune response involves lymphocytes and other leukocytes
- ▶ Cytokines play key role in the complex interaction between cells of the Immune system



Cytokines

- ◆ Cytokines secreted by some leukocytes and acting on other leukocytes are called **interleukins**
- ◆ Cytokine secreted by lymphocytes are called **lymphokines**
- ◆ Cytokines secreted by monocytes and macrophages are sometimes called **monokines**
- ◆ Cytokines that cause chemotaxis of leukocytes are called **chemokines**

Cytokines



- ◆ Autocrine: Acts on the same cell that secretes it
- ◆ Paracrine: Acts on adjacent cells
- ◆ Endocrine: Carried by the blood or body fluid to a distant site and acts on distant cells

Cytokines

- ♦ One cytokine may act on various cells and produce various effects
- ♦ Many cytokines may act on the same cell and produce same effect
- ♦ One cytokine may increase action of the other
- ♦ One cytokine may inhibit action of the others

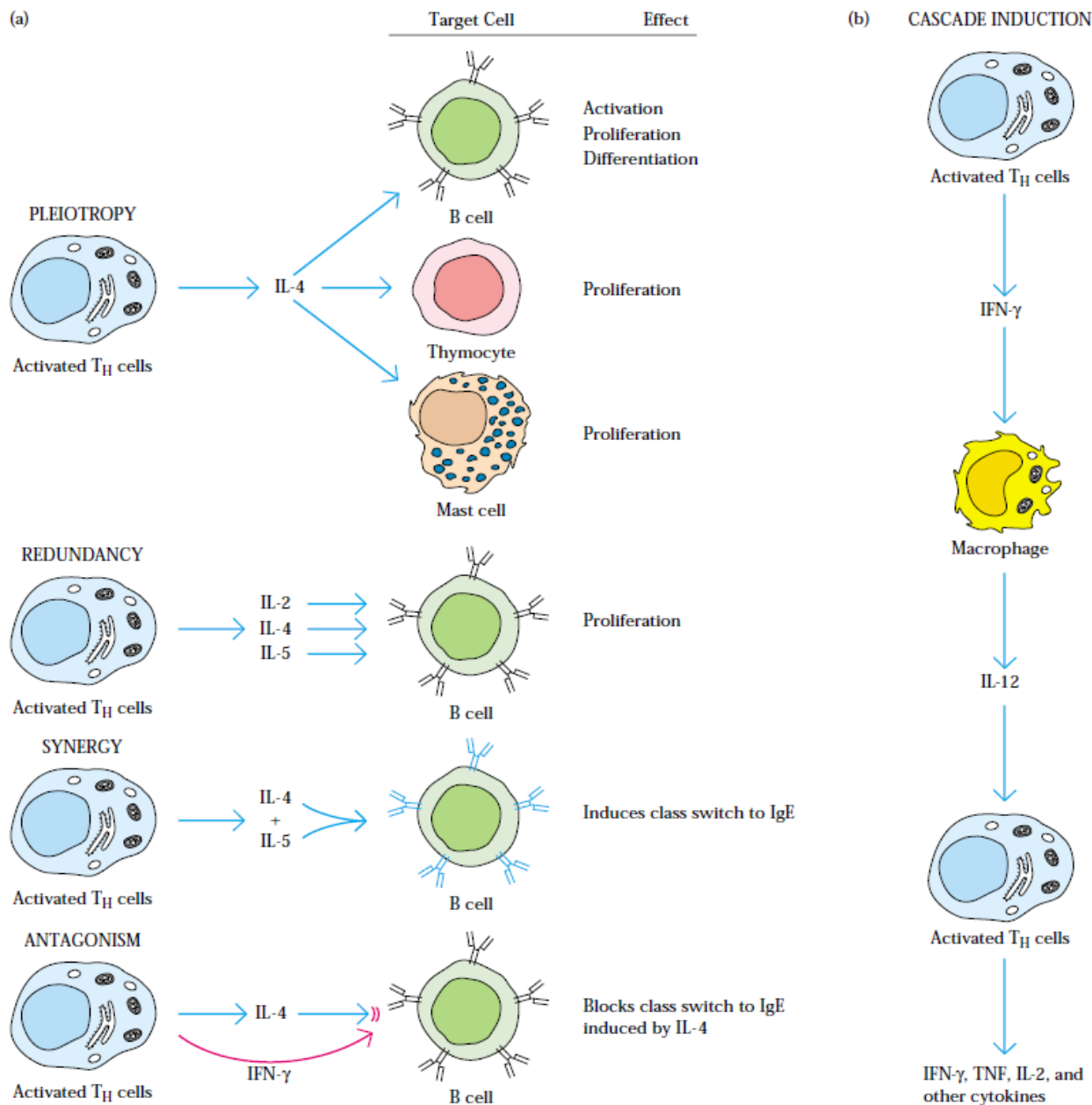


FIGURE 12-2 Cytokine attributes of (a) pleiotropy, redundancy, synergy (synergism), antagonism, and (b) cascade induction.

TABLE 12-1

Functional groups of selected cytokines¹

Cytokine*	Secreted by**	Targets and effects
SOME CYTOKINES OF INNATE IMMUNITY		
Interleukin 1 (IL-1)	Monocytes, macrophages, endothelial cells, epithelial cells	Vasculature (inflammation); hypothalamus (fever); liver (induction of acute phase proteins)
Tumor Necrosis Factor- α (TNF- α)	Macrophages	Vasculature (inflammation); liver (induction of acute phase proteins); loss of muscle, body fat (cachexia); induction of death in many cell types; neutrophil activation
Interleukin 12 (IL-12)	Macrophages, dendritic cells	NK cells; influences adaptive immunity (promotes T _H 1 subset)
Interleukin 6 (IL-6)	Macrophages, endothelial cells	Liver (induces acute phase proteins); influences adaptive immunity (proliferation and antibody secretion of B cell lineage)
Interferon α (IFN- α) (This is a family of molecules)	Macrophages	Induces an antiviral state in most nucleated cells; increases MHC class I expression; activates NK cells
Interferon β (IFN- β)	Fibroblasts	Induces an antiviral state in most nucleated cells; increases MHC class I expression; activates NK cells
SOME CYTOKINES OF ADAPTIVE IMMUNITY		
Interleukin 2 (IL-2)	T cells	T-cell proliferation; can promote AICD. NK cell activation and proliferation; B-cell proliferation
Interleukin 4 (IL-4)	T _H 2 cells; mast cells	Promotes T _H 2 differentiation; isotype switch to IgE
Interleukin 5 (IL-5)	T _H 2 cells	Eosinophil activation and generation
Interleukin 25 (IL-25)	Unknown	Induces secretion of T _H 2 cytokine profile
Transforming growth factor β (TGF- β)	T cells, macrophages, other cell types	Inhibits T-cell proliferation and effector functions; inhibits B-cell proliferation; promotes isotype switch to IgE; inhibits macrophages
Interferon γ (IFN- γ)	T _H 1 cells; CD8 ⁺ cells; NK cells	Activates macrophages; increases expression MHC class I and class II molecules; increases antigen presentation

¹Many cytokines play roles in more than one functional category.

*Only the major cell types providing cytokines for the indicated activity are listed; other cell types may also have the capacity to synthesize the given cytokine.

**Also note that activated cells generally secrete greater amounts of cytokine than unactivated cells.

Assignment

- ✓ Subset of CD 4 T cells
- ✓ Migration of Naïve and Memory T cells
- ✓ Cytokines